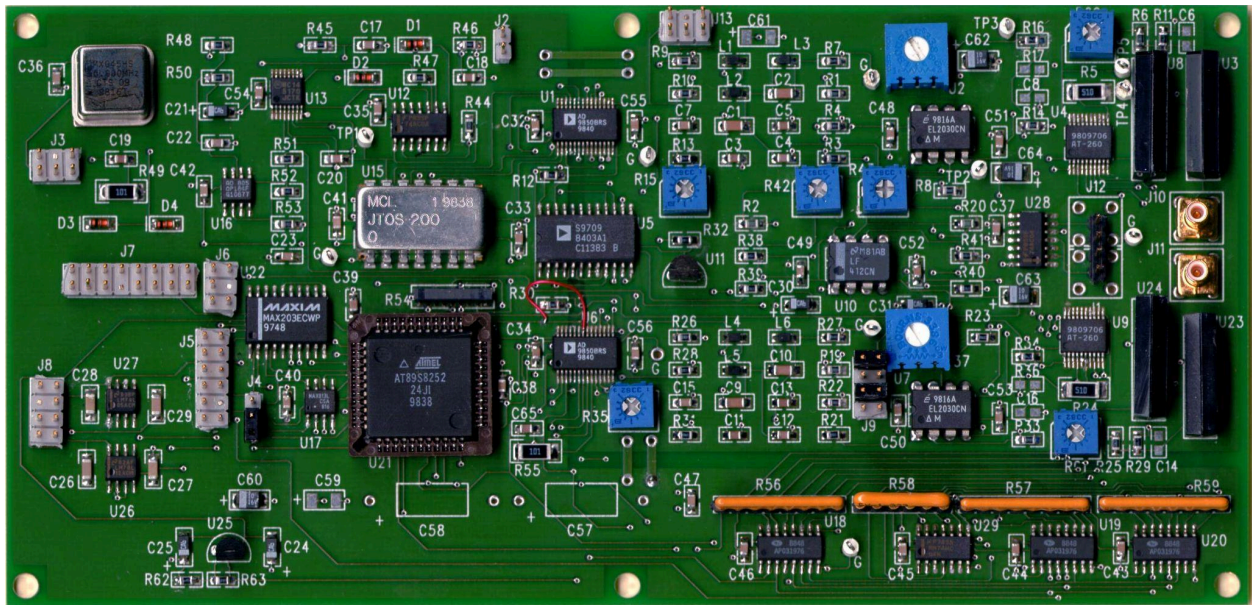
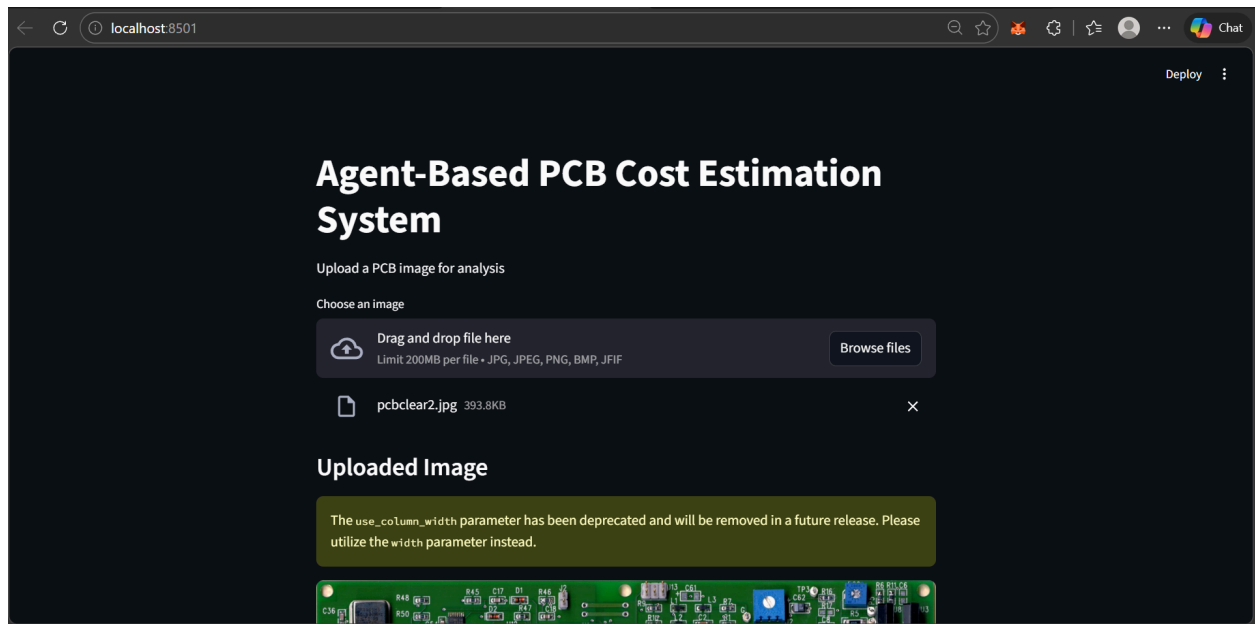


Input image pcbclear2.jpg:



Output of [app.py](#) (Streamlit UI):



localhost:8501

Deploy

Analysis

Time taken: 404.21 seconds

Agent completed in 3 step(s)

BOM Estimation Result

Estimated BOM Table

	component	estimated_count	unit_cost_inr	estimated_total_inr
0	resistor	35	0.5-5	17.5-175
1	capacitor	39	1-50	39-1950
2	IC	21	20-500+	420-10500+

Full LLM Output

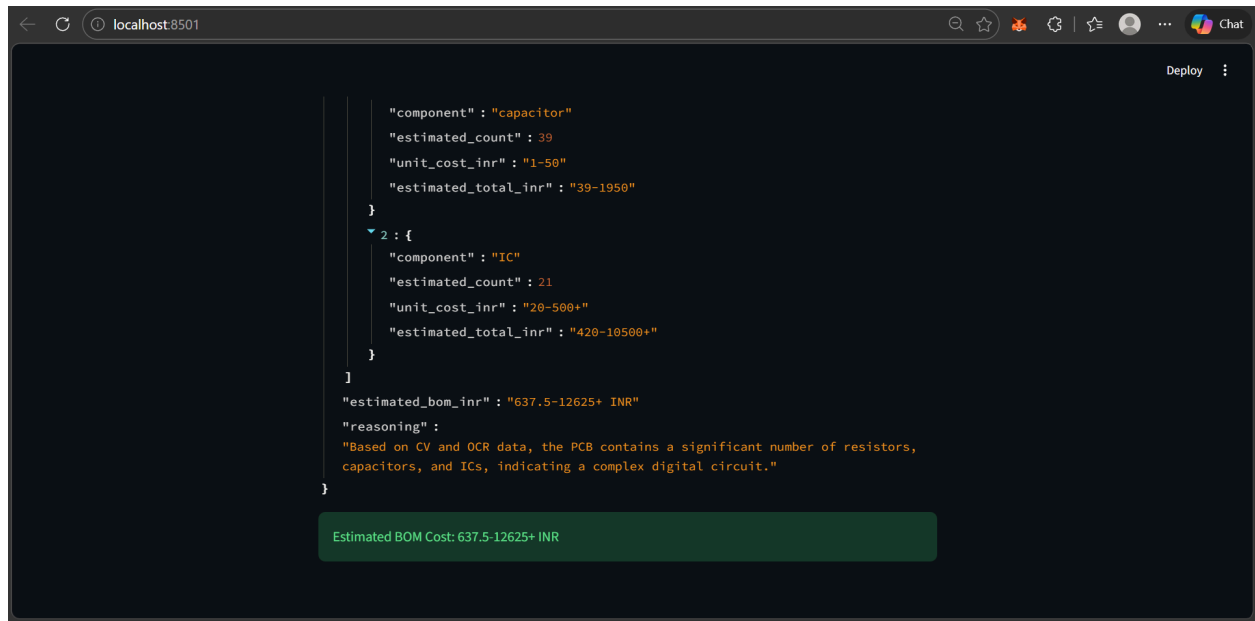
```
{
  "complexity": "High"
  "pcb_type": "Complex Digital PCB"
}
```

localhost:8501

Deploy

Full LLM Output

```
{
  "complexity": "High"
  "pcb_type": "Complex Digital PCB"
  "bom_table": [
    0: {
      "component": "resistor"
      "estimated_count": 35
      "unit_cost_inr": "0.5-5"
      "estimated_total_inr": "17.5-175"
    }
    1: {
      "component": "capacitor"
      "estimated_count": 39
      "unit_cost_inr": "1-50"
      "estimated_total_inr": "39-1950"
    }
    2: {
      "component": "IC"
      "estimated_count": 21
      "unit_cost_inr": "20-500+"
    }
  ]
}
```



```

{
  "component": "capacitor",
  "estimated_count": 39,
  "unit_cost_inr": "1-50",
  "estimated_total_inr": "39-1950"
}
2 : {
  "component": "IC",
  "estimated_count": 21,
  "unit_cost_inr": "20-500+",
  "estimated_total_inr": "420-10500+"
}
}
"estimated_bom_inr": "637.5-12625+ INR"
"reasoning":
"Based on CV and OCR data, the PCB contains a significant number of resistors,
capacitors, and ICs, indicating a complex digital circuit."
}

```

Estimated BOM Cost: 637.5-12625+ INR

Output of [main.py](#):

```

--- LLM Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'Complex Digital Circuit', '
bom_table': [{'component': 'resistor', 'estimated_count': 33, 'unit_cost_inr': '
0.5-5', 'estimated_total_inr': '16.5-165'}, {'component': 'capacitor', 'estimate
d_count': 39, 'unit_cost_inr': '1-50', 'estimated_total_inr': '39-1950'}, {'comp
onent': 'IC', 'estimated_count': 21, 'unit_cost_inr': '20-500+', 'estimated_tota
l_inr': '420-10500+'}], 'estimated_bom_inr': '637.5-12655.5 INR', 'reasoning': '
Based on OCR and CV data, the PCB contains a high number of components with sign
ificant area coverage, indicating a complex digital circuit.'}]

--- Reflection Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'Complex Digital Circuit', '
bom_table': [{'component': 'resistor', 'estimated_count': 35, 'unit_cost_inr': '
0.5-5', 'estimated_total_inr': '17.5-175'}, {'component': 'capacitor', 'estimate
d_count': 39, 'unit_cost_inr': '1-50', 'estimated_total_inr': '39-1950'}, {'comp
onent': 'IC', 'estimated_count': 21, 'unit_cost_inr': '20-500+', 'estimated_tota
l_inr': '420-10500+'}], 'estimated_bom_inr': '676.5-12625.5 INR', 'reasoning': '
Based on the component counts derived from CV data, the PCB contains a high numb
er of components with significant area coverage, indicating a complex digital ci
rcuit.'}]

BOM TABLE
-----
resistor      | count=35      | unit=0.5-5      | total=17.5-175
capacitor     | count=39      | unit=1-50       | total=39-1950
IC            | count=21      | unit=20-500+    | total=420-10500+
-----

Estimated BOM Cost: 676.5-12625.5 INR

Full Result:
{'complexity': 'High', 'pcb_type': 'Complex Digital Circuit', 'bom_table': [{'co
mponent': 'resistor', 'estimated_count': 35, 'unit_cost_inr': '0.5-5', 'estimate
d_total_inr': '17.5-175'}, {'component': 'capacitor', 'estimated_count': 39, 'un
it_cost_inr': '1-50', 'estimated_total_inr': '39-1950'}, {'component': 'IC', 'es
timated_count': 21, 'unit_cost_inr': '20-500+', 'estimated_total_inr': '420-1050
0+'}], 'estimated_bom_inr': '676.5-12625.5 INR', 'reasoning': 'Based on the comp
onent counts derived from CV data, the PCB contains a high number of components
with significant area coverage, indicating a complex digital circuit.'}]
(newvenv)
HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version6 (main)

```

Output of [ocr.py](#):

```

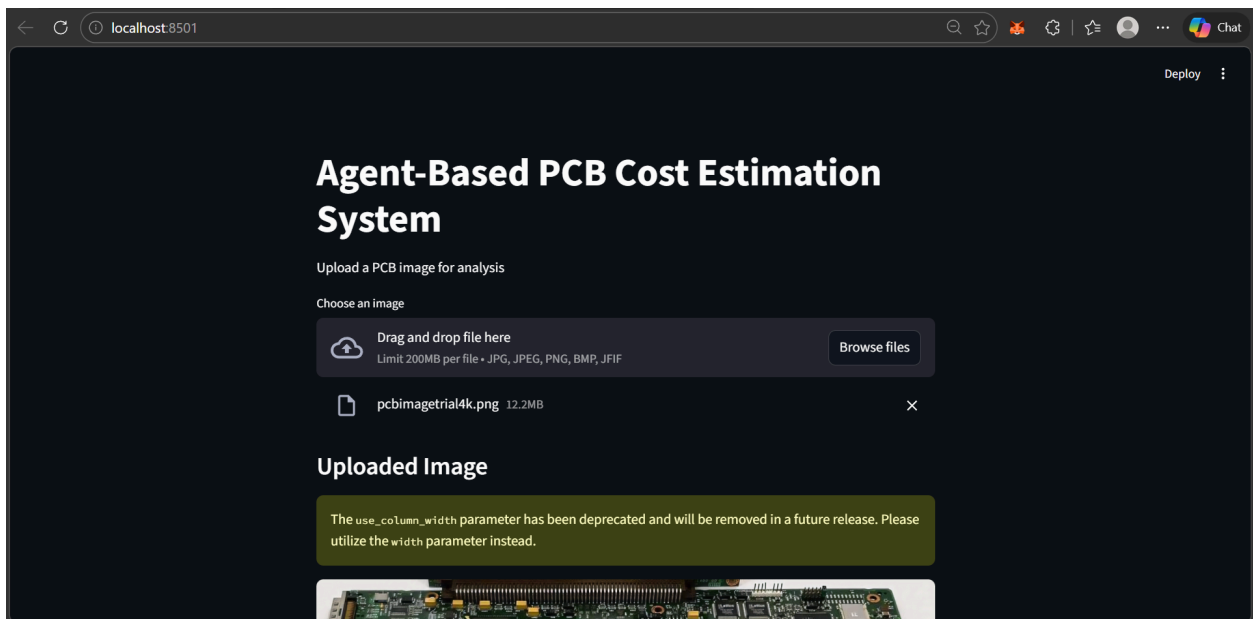
HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version6 (main)
$ python ocr.py
--- LOADING IMAGE ---
--- RUNNING OCR ---
--- OCR RAW TEXT (first 50) ---
['R6', 'R11C6', 'R45', 'C17', 'D1', 'R46', 'TP30', 'R16', 'R48', 'L', 'C62', 'C36', 'R47', 'C18', 'F1', 'R50', 'HHIN', 'R5', 'U8', 'U3', 'C54', 'U13', 'U12', 'III', 'R4
4', 'U1m', 'Iiit!', 'C55', 'C21+s', 'iiiiio', 'C35', 'C320', 'BR', 'C48', '9816A', 'C519', 'KL', 'J3', 'C19', 'C22', 'GD', 'R51', 'TPIC', 'MNI11', '21', 'EL2030CN', 'UE', '
C64', 'R49', 'C42,']
--- OCR REFERENCE COUNTS ---
{'R': 33, 'C': 34, 'U': 20, 'J': 6}
--- OCR IC CANDIDATES ---
['TP30', 'C18', '9816A', 'EL2030CN', 'CM81AB', 'L22', 'R35', '12CN', 'MAX203ECWP', 'UNIMMIOC56', 'CIQ', 'AT89S8252', '24JI', '69816A', 'EL2030CN', 'JC29', 'OR59', 'DMHU29',
'R62EE', 'C4SILIJJIII']
--- DONE ---
(newvenv)
HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version6 (main)
$ |

```

Input image pcbimagetrial4k.png:



Output of [app.py](#) (Streamlit UI):



localhost:8501

Deploy

Analysis

Time taken: 372.90 seconds

Agent completed in 3 step(s)

BOM Estimation Result

Estimated BOM Table

	component	estimated_count	unit_cost_inr	estimated_total_inr
0	resistor	808	0.5-5	404-4040 INR
1	IC	42	20-500+	840-21000 INR

Full LLM Output

```
{  "complexity": "High"  "pcb_type": "Complex Digital Circuit"  "bom_table": [    {      "component": "resistor"      "estimated_count": 808      "unit_cost_inr": "0.5-5"      "estimated_total_inr": "404-4040 INR"    }    {      "component": "IC"      "estimated_count": 42      "unit_cost_inr": "20-500+"      "estimated_total_inr": "840-21000 INR"    }  ]  "estimated_bom_inr": "844-25040 INR"  "reasoning": "Based on component counts and IC presence, the board is likely a complex digital circuit with a high number of resistors (808) and multiple potential IC chips (42)."}  }
```

Estimated BOM Cost: 844-25040 INR

Output of [main.py](#):


```

--- LLM Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'Customized Embedded System', 'bom_table': [{'component': 'IC', 'estimated_count': 1, 'unit_cost_inr': '20-500+ INR', 'estimated_total_inr': '20-500+ INR'}, {'component': 'resistor', 'estimated_count': 849, 'unit_cost_inr': '0.5-5 INR', 'estimated_total_inr': '424.5-4245 INR'}, {'component': 'capacitor', 'estimated_count': 0, 'unit_cost_inr': '1-50 INR', 'estimated_total_inr': '0-0 INR'}], 'estimated_bom_inr': '20-4745+ INR', 'reasoning': 'Based on CV analysis, the board has a high component count with primarily resistors and ICs. The OCR provided one possible IC name but did not confirm exact details.'}}

--- Reflection Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'Customized Embedded System', 'bom_table': [{'component': 'IC', 'estimated_count': 42, 'unit_cost_inr': '20-500+ INR', 'estimated_total_inr': '840-21000+ INR'}, {'component': 'resistor', 'estimated_count': 808, 'unit_cost_inr': '0.5-5 INR', 'estimated_total_inr': '404-4040 INR'}], 'estimated_bom_inr': '840-25100+ INR', 'reasoning': 'CV analysis provides a detailed component count, with 808 resistors and 42 ICs. OCR did not provide specific IC names but confirmed the presence of at least one IC.'}}

BOM TABLE
-----
IC          | count=42 | unit=20-500+ INR | total=840-21000+ INR
resistor    | count=808 | unit=0.5-5 INR   | total=404-4040 INR
-----

Estimated BOM Cost: 840-25100+ INR

Full Result:
{'complexity': 'High', 'pcb_type': 'Customized Embedded System', 'bom_table': [{'component': 'IC', 'estimated_count': 42, 'unit_cost_inr': '20-500+ INR', 'estimated_total_inr': '840-21000+ INR'}, {'component': 'resistor', 'estimated_count': 808, 'unit_cost_inr': '0.5-5 INR', 'estimated_total_inr': '404-4040 INR'}], 'estimated_bom_inr': '840-25100+ INR', 'reasoning': 'CV analysis provides a detailed component count, with 808 resistors and 42 ICs. OCR did not provide specific IC names but confirmed the presence of at least one IC.'}}
(newvenv)
HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version6 (main)
$ |

```

Output of [ocr.py](#):

```

HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version6 (main)
$ python ocr.py

--- LOADING IMAGE ---

--- RUNNING OCR ---

--- OCR RAW TEXT (first 50) ---
['ut', 'RN', 'i:', 'FTXTXTEEEI', 'VCHI', 'a1', 'WI', 'mE', 'IEE', 'Fiitiiif', 'iliiiiii', 'Smm', 'iiniifs', 'MADE', 'IN', 'ETAIWAN', 'G7F']

--- OCR REFERENCE COUNTS ---
{'R': 0, 'C': 0, 'U': 0, 'J': 0}

--- OCR IC CANDIDATES ---
['G7F']

--- DONE ---
(newvenv)
HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version6 (main)
$ |

```